Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

A drop of water of volume V is pressed between two glass plates so that it spreads to an area A. If T is surface tension, then the normal force required to separate the glass plates is

(A) $\frac{TA^2}{V}$

(B) $\frac{TA^2}{2V}$

(C) $\frac{2TA^2}{V}$

(D) $\frac{4TA^2}{V}$

Q.2

Two liquids, ρ_1 and ρ_2, are in equilibrium in a U-tube that is open at both ends. The liquids do not mix, and liquid 1 rests on top of liquid 2. How is the density ρ_1 of liquid 1 related to the density ρ_2 of liquid 2?

(A) ρ_1 is equal to ρ_2, since the liquids are in equilibrium.

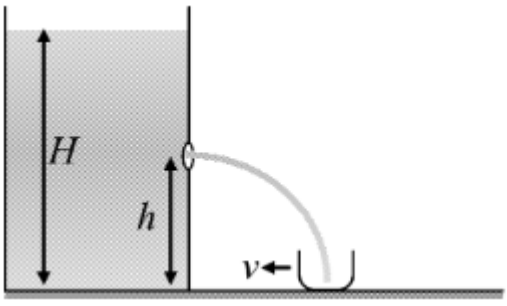
(B) ρ_1 is greater than ρ_2

(C) ρ_1 is less than ρ_2.

(D) There is not enough information to tell which liquid has the greater density.

Q.3

A liquid is filled in a container of area A upto a height H. A small hole of area a is made at a height h. A beaker (very small area) is moving on the surface. The speed with which beaker must move so that liquid always falls into it.



(A) $\sqrt{2gh}$

(B) $\frac{A}{a}\sqrt{2g(H-h)}$

(C) $\frac{a}{A}\sqrt{2gh}$

(D) $\sqrt{2g(H-h)}$

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Q.4

A cube of iron of edge $5\ \text{cm}$ floats on the surface of mercury, contained in a tank. Water is then poured on top, so that the cube just gets immersed. The depth of cube in water layer is:
(Specific gravities of iron and mercury are 8.56 and 13.6, respectively.)

(A) $1\ \text{cm}$

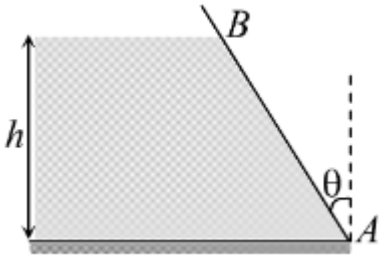
(B) $1.8\ \text{cm}$

(C) $2\ \text{cm}$

(D) $2.2\ \text{cm}$

Q.5

The height of water in a vessel is h. The vessel wall of width b is at an angle θ to the vertical. The net force exerted by the water on the wall is



(A) $\frac{1}{3}\rho bgh^2g\cos\theta$

(B) $bh^2\rho g$

(C) $\frac{1}{2}\rho ph^2g\sec\theta$

(D) zero

Q.6

A block of material of specific gravity 0.4 is held submerged at a depth of $1\ \text{m}$ in a vessel filled with water. The vessel is accelerated upwards with acceleration of $a_0 = \text{g}/5$. If the block is released at $t = 0\ \text{s}$, neglecting viscous effects, it will reach the water surface at t equal to $\left(g - 10\ \text{m/s}^2\right)$

(A) $0.60\ \text{s}$

(B) $0.33\ \text{s}$

(C) $3.3\ \text{s}$

(D) $1.2\ \text{s}$

Q.7

$0.4\ \text{mm}\&0.6\ \text{mm}$ Diameter tubes form a U tube. What will be the difference between water level? Surface tension $\text{T} = 60$ Dyne $/\text{cm}$

(A) $2\ \text{cm}$

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(B) $2.3\ \text{cm}$

(C) $3\ \text{cm}$

(D) $2.04\ \text{cm}$

Q.8

Water rises in a capillary tube upto a height of $10\ \text{cm}$ where as mercury depresses in it by $3.42\ \text{cm}$. If the angle of contact and density of mercury are 135° and $13.6\text{gm}/\text{cc}$ respectively then the ratio of the surface tensions of water and mercury will be -

(A) $6.5 : 1$

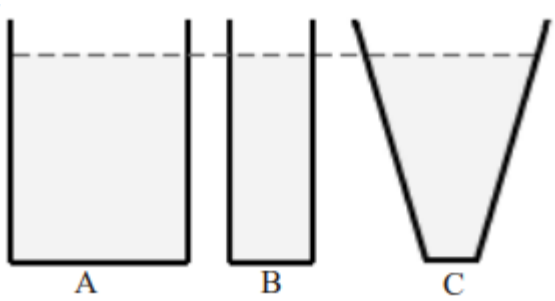
(B) $1 : 6.5$

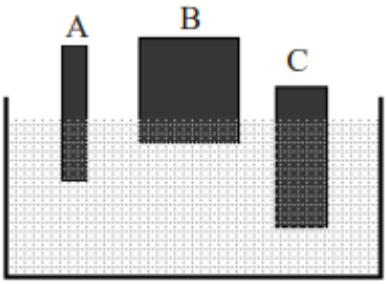
(C) $1 : 5.6$

(D) $5.6 : 1$

Q.9

Choose the correct statements -





(A) The figure shows three containers filled to the same height with the same fluid. All three contains have the same pressure at the bottom.

(B) In above figure container A has more pressure than B at the bottom.

(C) Three solid objects are floating in a liquid. They have different weights and volumes then correct rank for the objects according to their density, largest first is $\text{C}, \text{A}, \text{B}$.

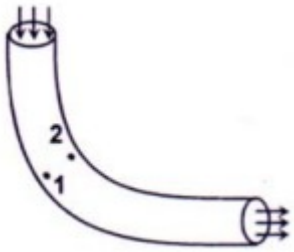
(D) In above figure the correct rank for the objects according to their density, largest first is A, C, B.

Q.10

Ideal fluid flows along a tube of uniform cross section, located in a horizontal plane and bent as shown in figure. The flow is steady, 1 and 2 are two points in the tube. If P_1 and P_2 are pressure at the two points and v_1 and v_2 are the

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respective velocities, then



(A) $P_1 < P_2$

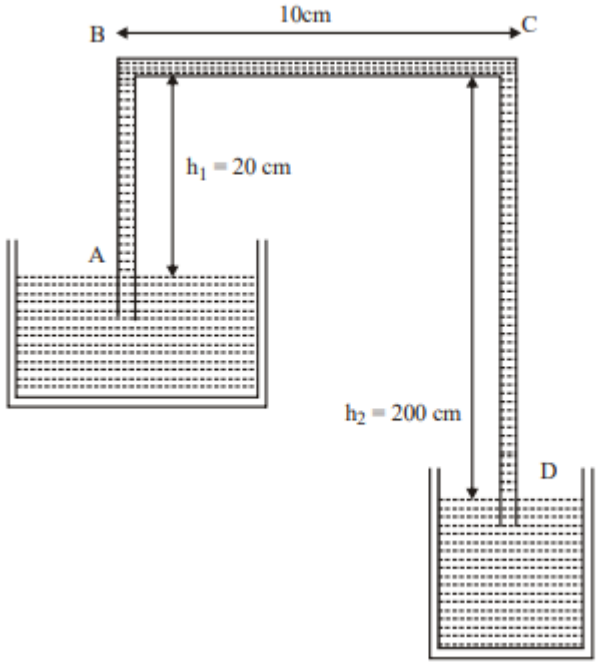
(B) $P_1 > P_2$

(C) $\mathrm{v}_1 < \mathrm{V}_2$

(D) $v_1 > v_2$

Q.11

Figure shows a siphon in action. The liquid flowing through the tube has a density of $1.5\ \mathrm{g/cc}$. $\left(\mathrm{g} = 980\ \mathrm{cm/s^2}\right)$



(A) pressure difference between points B and C is $2.65 \times 10^4\ \mathrm{N/m^2}$.

(B) pressure difference between B and C is independent of length BC.

(C) velocity of efflux is independent of the pressure at C.

(D) velocity of efflux is independent of the pressure gradient across BC.

Q.12

Passage:

Bernoulli's equation is nothing but a statement about the conservation of mechanical energy in a system. From work energy theorem we know that work done by all the forces is equal to change in kinetic energy. $\Delta \mathrm{w} = \Delta \mathrm{k}$ or $\Delta \mathrm{w_c} + \Delta \mathrm{w_{nc}} = \Delta \mathrm{k}$

where, $\Delta \mathrm{w_c}$ is work done by conservative forces and $\Delta \mathrm{w_{nc}}$ is work done by non-conservative forces. But in case of conservative field work done by conservative force is equal to negative of the change in potential energy of the system so $\Delta \mathrm{w_c} = -\Delta \mathrm{u}$ or $\Delta \mathrm{w_c} = \Delta \mathrm{k} + \Delta \mathrm{v}$

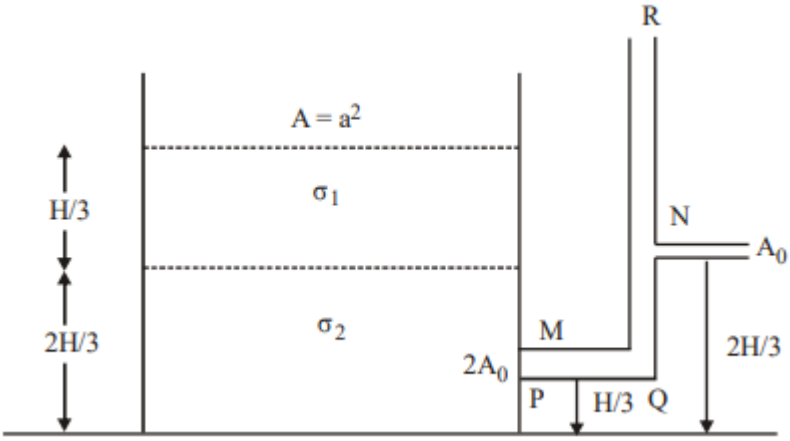
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So, in the Bernoulli's equation these three terms refers to the work done by the pressure force, change in kinetic energy and change in potential energy. All quantities being taken as per unit volume of the non-viscous and incompressible fluid.

So, the equation is, $P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$.

Here we only take gravitational potential energy.

Question: The cross-sectional area of the beaker is square of side a and that of two holes, M and N are $2\ \mathrm{A}_0$ and A_0. Ratio of densities of liquids is $\frac{\sigma_1}{\sigma_2} = \frac{1}{2}$. Cross-sectional area of the pipe PQR is uniform. Initially holes are closed and both fluids are incompressible and non-viscous. If hole at M only is opened then the height of the liquid in pipe QR is -



(A) $\mathrm{H}/3$

(B) $2\mathrm{H}/3$

(C) $\mathrm{H}/2$

(D) is not constant

Q.13

Passage:

Bernoulli's equation is nothing but a statement about the conservation of mechanical energy in a system. From work energy theorem we know that work done by all the forces is equal to change in kinetic energy. $\Delta \mathrm{w} = \Delta \mathrm{k}$ or $\Delta \mathrm{w_c} + \Delta \mathrm{w_{nc}} = \Delta \mathrm{k}$

where, $\Delta \mathrm{w_c}$ is work done by conservative forces and $\Delta \mathrm{w_{nc}}$ is work done by non-conservative forces. But in case of conservative field work done by conservative force is equal to negative of the change in potential energy of the system so $\Delta \mathrm{w_c} = -\Delta \mathrm{u}$ or $\Delta \mathrm{w_c} = \Delta \mathrm{k} + \Delta \mathrm{v}$

So, in the Bernoulli's equation these three terms refers to the work done by the pressure force, change in kinetic energy and change in potential energy. All quantities being taken as per unit volume of the non-viscous and incompressible fluid.

So, the equation is, $P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$.

Here we only take gravitational potential energy.

Question: The velocity of the fluid at N after a small time of opening both the holes (you assume that the height of liquids in beaker in this time interval does not change)

(A) $\sqrt{\frac{\mathrm{gH}}{3}}$

(B) $\sqrt{2\ \mathrm{g}\frac{\mathrm{H}}{3}}$

(C) $\sqrt{\mathrm{g}\frac{\mathrm{H}}{6}}$

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(D) $\frac{1}{3}\sqrt{\frac{gH}{4}}$

Q.14

Passage:

Bernoulli's equation is nothing but a statement about the conservation of mechanical energy in a system. From work energy theorem we know that work done by all the forces is equal to change in kinetic energy. $\Delta w = \Delta k$ or $\Delta w_c + \Delta w_{nc} = \Delta k$

where, Δw_c is work done by conservative forces and Δw_{nc} is work done by non-conservative forces. But in case of conservative field work done by conservative force is equal to negative of the change in potential energy of the system so $\Delta w_c = -\Delta u$ or $\Delta w_c = \Delta k + \Delta v$

So, in the Bernoulli's equation these three terms refers to the work done by the pressure force, change in kinetic energy and change in potential energy. All quantities being taken as per unit volume of the non-viscous and incompressible fluid.

So, the equation is, $P + \frac{1}{2}\rho v^2 + \rho g h =$ constant.

Here we only take gravitational potential energy.

Question: Initial force on the beaker exerted by water efflux is -

(A) $\frac{2\sigma_2 A_0 gH}{3}$

(B) $\frac{\sigma_2 A_0 gH}{3}$

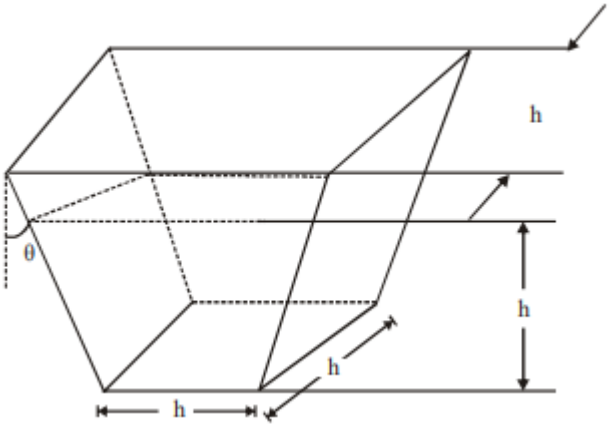
(C) $\sigma_2 A_0 g\frac{H}{6}$

(D) $\sigma_2 A_0 \frac{H}{2}$

Q.15

Passage:

A vessel as shown above has a square base of side h, two rectangular sides and two trapezoidal sides. The rectangular side is inclined to the vertical by angle θ as shown. The vessel has liquid to a height h.



If F_b, F_r, F_t are the hydrostatic thrust on the base, a rectangular side and a trapezoidal side respectively.

Question: For $\theta = 0°$. Choose the correct option -

(A) $F_r = \frac{F_b}{\sqrt{3}}$

(B) $F_t = \frac{F_b}{2}$

(C) $F_r = \frac{F_b}{\sqrt{2}}$

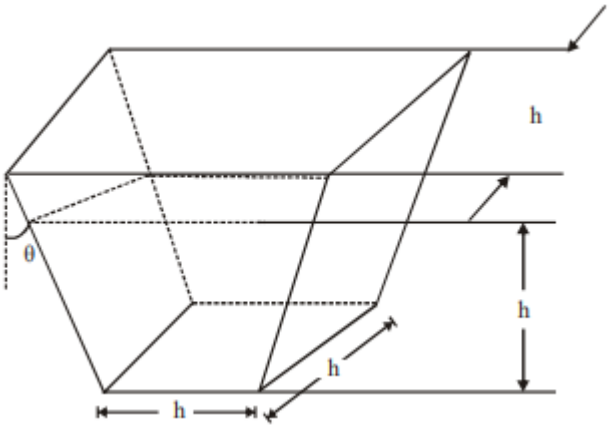
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(D) None of these

Q.16

Passage:

A vessel as shown above has a square base of side h, two rectangular sides and two trapezoidal sides. The rectangular side is inclined to the vertical by angle θ as shown. The vessel has liquid to a height h.



If $\mathrm{F_b}, \mathrm{F_r}, \mathrm{F_t}$ are the hydrostatic thrust on the base, a rectangular side and a trapezoidal side respectively.

Question: For $\theta = 30^\circ$. Choose the correct option -

(A) $\mathrm{F_r} = \frac{\mathrm{F_b}}{\sqrt{3}}$

(B) $\mathrm{F_t} = \frac{\mathrm{F_b}}{2}$

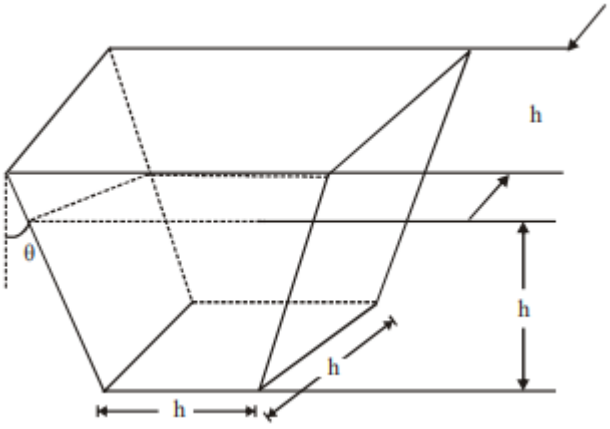
(C) $\mathrm{F_r} = \frac{\mathrm{F_b}}{\sqrt{2}}$

(D) None of these

Q.17

Passage:

A vessel as shown above has a square base of side h, two rectangular sides and two trapezoidal sides. The rectangular side is inclined to the vertical by angle θ as shown. The vessel has liquid to a height h.



If $\mathrm{F_b}, \mathrm{F_r}, \mathrm{F_t}$ are the hydrostatic thrust on the base, a rectangular side and a trapezoidal side respectively.

Question: For $\theta = 45^\circ$. Choose the correct option -

(A) $\mathrm{F_r} = \frac{\mathrm{F_b}}{\sqrt{3}}$

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(B) $F_t = \frac{F_b}{2}$

(C) $F_r = \frac{F_b}{\sqrt{2}}$

(D) None of these

Q.18

Surface tension, density and angle of contact with glass for water, a liquid and Mercury are respectively as below :
Water: $T, \rho, 0^\circ$
Liquid : $0.5\ T, 0.8\rho, 37^\circ$
Mercury : $6.8\ T, 13.6\rho, 135^\circ$
If a given glass capillary is dipped in mercury, the capillary rise is $-1.4\ \text{cm.}$ and angle of contact is 135°
The same capillary is now dipped in either water or the liquid, to various depths.
Column I gives the resulting angle of contact. Column II gives possible liquid column height match them.

Column I	**Column II**
(A) 37°	(p) 1.5cm.
(B) 53°	(q) 2cm.
(C) 60°	(r) 3.2 cm
(D) 90°	(S) a value different from above

(A) A-(qr) B-(ps) C-(qs) D-(s)

(B) A-(qr) B-(p) C-(s) D-(s)

(C) A-(qr) B-(ps) C-(qsr) D-(s)

(D) A-(qr) B-(psr) C-(qs) D-(sr)

Q.19

A long capillary tube of radius 0.2 mm is placed vertically inside a beaker of water. Water rises to a hight of 7.05 cm in the tube making zero angle os contact. Now if tube is pushed into water so that only 5 cm of its length is above the surface, What is the new angle of contact in degrees between water and glass surface?(Take $\sqrt{2} = 1.41$, surface tension of water $0.07\ \text{N/m}$)

Q.20

In seawater, a life preserver with a volume of $0.0400\ \text{m}^3$ will support a $75.0 - \text{kg}$ person (average density $980\ \text{kg/m}^3$), with 20% of the person's volume above the water surface when the life preserver is fully submerged. What is the density of the material composing the life preserver?

Answers & Solutions

Q.1 Answer:

$\frac{2TA^2}{V}$

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Solution:

Excess pressure $P = \frac{T}{R}$

$$R = \frac{d}{2} \quad \therefore \quad P = \frac{2T}{d}$$

$$P = \frac{2TA}{V} \quad \left[\therefore d = \frac{V}{A}\right]$$

$$F = \frac{2TA^2}{V}$$

$$\text{Force} = \text{Pressure} \times \text{Area}$$

Q.2 Answer:

ρ_1 is less than ρ_2.

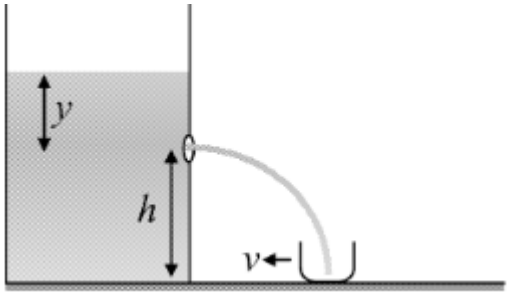
Solution:

The pressure at the top of each liquid is the same, since the U-tube is open at both ends. Also, the pressure at the location of the dashed line is the same in both the left and right sides of the U-tube, since these two locations are at the same level. Thus, the pressure increment ρ_1 **gh** $_1$ **for liquid 1 must be equal to the pressure increment** $\rho_2 g h_2$ **for liquid 2, where** h_1 **and** h_2 **are the heights of the liquids above the dashed line. Since** h_1 **is greater than** $\mathrm{h}_2, 1$ **must be less than 2 .**

Q.3 Answer:

$\frac{a}{A}\sqrt{2gh}$

Solution:



$$v = \sqrt{2gy}$$

$$R = \sqrt{\frac{2h}{g}} \times \sqrt{2gy} = 2\sqrt{h}\sqrt{y}$$

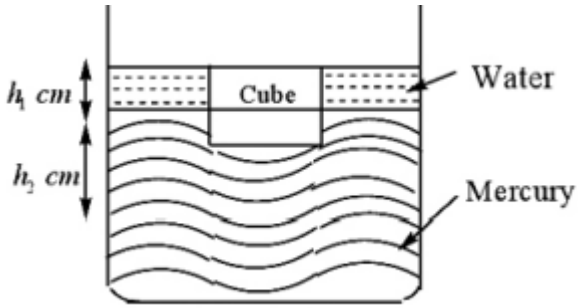
$$\frac{dR}{dt} = 2\sqrt{h}\frac{1}{2\sqrt{y}}\frac{dy}{dt} \quad a\sqrt{2gy} = A\frac{dy}{dt} \Rightarrow \frac{dR}{dt} = \frac{a}{A}\sqrt{2gh}$$

Q.4 Answer:

2 cm

Solution:

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$$h_2 = (5 - h_1)\,\text{cm}$$

Weight of water displaced $= \left(5^2 h_1\right) g$ **dyn and weight of mercury displaced** $5^2\,(5 - h_1)\,13.6$ g **dyn. Therefore, upthrust offered by water and mercury together is given by**

$$5^2 h_1 g + 5^2\,(5 - h_1)\,13.6g$$
$$= 5^2\,(68 - 12.6h_1)\,g dyn$$

Weight of cube $= 5^3 \times 8.56$ gdyn

From the principle of floatation, we have

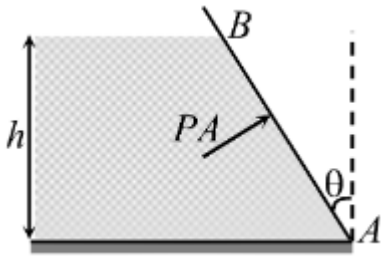
Upthrust = weight of the floating body

$$\therefore 5^2\,(68 - 12.6h_1)\,g = 5^3 \times 8.56g$$
$$\Rightarrow 68 - 12.6h_1 = 41.8 \Rightarrow h_1 = 25.2/(12.6) = 2\ \text{cm}$$

Q.5 Answer:

$\frac{1}{2}\rho p h^2 g \sec\theta$

Solution:



Average pressure on the wall

$= \frac{\rho g h}{2}$

$AB = \frac{h}{\cos\theta}$

Therefore area of the wall $= \frac{h}{\cos\theta} b$

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Net force exerted by the water on the wall

$= PA = \frac{\rho g h}{2} \frac{h}{\cos\theta} b = \frac{1}{2} \rho b h^2 g \sec\theta$

Q.6 Answer:

0.33 s

Solution:

In frame of vessel

$$-V\rho_s\left(g+\frac{g}{5}\right)+V\rho_\ell\left(g+\frac{g}{5}\right)=V\rho_s a$$

$$\frac{(\rho_\ell-\rho_s)}{\rho_s}\left(g+\frac{g}{5}\right)=a$$

$$\frac{0.6}{0.4}(12)=a$$

$$a=18\ \mathrm{m/s^2}$$

$$s=ut+\frac{1}{2}at^2$$

$$1=\frac{1}{2}\times 18t^2$$

$$t^2=\frac{1}{9}\Rightarrow t=\frac{1}{3}s$$

Q.7 Answer:

2.04 cm

Solution:

Let Angle of contact is zero for glass & water.

For first tube, $\mathrm{h_1}\rho g = \frac{4\,\mathrm{T}}{\mathrm{d_1}}\cos 0^\circ = \frac{4\,\mathrm{T}}{\mathrm{d_1}}$ **for second tube,** $\mathrm{h_2}\rho g = \frac{4\,\mathrm{T}}{\mathrm{d_2}}\cos 0^\circ = \frac{4\,\mathrm{T}}{\mathrm{d_2}}\ \rho g\,(h_1-h_2) = 4T\left[\frac{1}{d_1}-\frac{1}{d_2}\right]$

$\mathrm{h_1}-\mathrm{h_2} = \frac{4\times 60}{1\times 981}\left[\frac{1}{0.04}-\frac{1}{0.06}\right] = 2.038 = 2.04$ cm

Q.8 Answer:

$1:6.5$

Solution:

The height h **through which a liquid will rise in a capillary tube of radius** r **is given by** $\mathrm{h} = \frac{2\,\mathrm{S}\cos\theta}{\mathrm{r}\rho g}$ **where** S **is the surface tension,** ρ **is the density of the liquid and** θ **is the angle of contact.**

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For identical capillary tubes and two liquids (water '1' \& mercury '2')

$$\frac{\mathrm{h}_1}{\mathrm{h}_2}=\frac{2\,\mathrm{S}_1\cos\theta_1}{\mathrm{r}\rho_1\,\mathrm{g}}\times\frac{\mathrm{r}\rho_2\,\mathrm{g}}{2\,\mathrm{S}_2\cos\theta_2}$$

$$\Rightarrow\frac{\mathrm{h}_1}{\mathrm{h}_2}=\frac{\mathrm{S}_1\cos\theta_1}{\rho_1}\times\frac{\rho_2}{\mathrm{S}_2\cos\theta_2}$$

$$\Rightarrow\frac{\mathrm{S}_1}{\mathrm{S}_2}=\frac{\mathrm{h}_1}{\mathrm{h}_2}\times\frac{\cos\theta_2}{\cos\theta_1}\times\frac{\rho_1}{\rho_2}$$

$$\Rightarrow\frac{\mathrm{S}_1}{\mathrm{S}_2}=\frac{10}{-3.42}\times\frac{\cos 135^\circ}{\cos 0^\circ}\times\frac{1}{13.6}$$

$$\Rightarrow\frac{\mathrm{S}_1}{\mathrm{S}_2}=\frac{10}{-3.42}\times\frac{-0.707}{1}\times\frac{1}{13.6}$$

$$\Rightarrow\frac{\mathrm{S}_1}{\mathrm{S}_2}=0.152\approx\frac{1}{6.5}$$

Q.9 Answer:

The figure shows three containers filled to the same height with the same fluid. All three contains have the same pressure at the bottom.

Solution:

(1) According to the relation $\mathrm{P}_2=\mathrm{P}_1+\rho gh$, the pressure P_2 at the bottom of the container depends on the height h of the fluid above it.

Since this height is the same for all three containers, the pressure at the bottom is the same for each container.

(3) When an object floats, its weight ($\rho_{\text{object}}\,\mathrm{V}_{\text{object}}\,\mathrm{g}$) equals the buoyant force ($\rho_{\text{fluid}}\,\mathrm{V}_{\text{displaced}}\,\mathrm{g}$), where $V_{\text{displaced}}$ is the volume of fluid displaced by the object.

Thus, $\rho_{\text{object}}\,\mathrm{V}_{\text{object}}\,\mathrm{g}=\rho_{\text{fluid}}\,\mathrm{V}_{\text{displaced}}\,\mathrm{g}$, so the density of the object is

$\rho_{\text{object}}=\rho_{\text{fluid}}\left(\mathrm{V}_{\text{displaced}}/\mathrm{V}_{\text{object}}\right)$.

Thus, the density of the object is proportional to the ratio $\mathrm{V}_{\text{displaced}}/\mathrm{V}_{\text{object}}$ of the volumes.

This ratio is greatest for object C and least for B.

Q.10 Answer:

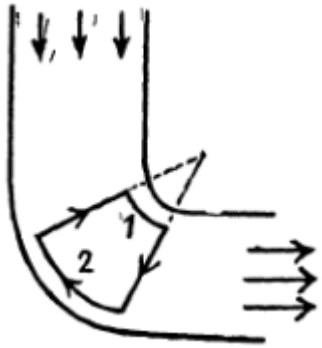
$P_1>P_2$

Solution:

Between 1 and 2 fluid particles are in nearly circular motion and therefore have centripetal acceleration. The force for this acceleration, like for any other situation in an ideal fluid, can only come from the pressure variation along the line joining 1 and 2. This requires that pressure at 1 should be greater than the pressure at 2 i.e.

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$P_1 > P_2$



so that the fluid particles can have required acceleration. If there is no turbulence, the motion can be taken as irrotational. Then by considering

$\oint \vec{v} \cdot dr = 0$

along the circuit shown we infer that

$v_2 > v_1$

(The portion of the circuit near 1 and 2 are streamlines while the other two arms are at right angle to streamlines)

Q.11 Answer:

pressure difference between points B and C is $2.65 \times 10^4\ N/m^2$.

Solution:

As points A and D are open to the atmosphere $p_A = p_D = p_0$

If p_B is the pressure at B, $p_A = p_0 = p_B + h_1 \rho g$

$\therefore \quad p_B = p_0 - h_1 \rho g$

Similarly, $p_C = p_0 - h_2 \rho g$

$$\therefore \quad p_B - p_C = (h_2 - h_1)\,\rho g = (200 - 20) \times 1.5 \times 980 = 2.65 \times 10^5 \text{dyne/cm}^2$$
$$= 2.65 \times 10^4\ N/m^2$$

Since $p_B - p_C \propto (h_2 - h_1)$, pressure difference between B and C is independent of length BC.

Velocity of efflux $v = \sqrt{2gh} = \sqrt{2\,g\,(\,h_2 - h_1)}$

Hence, velocity of efflux is dependent on the pressure at C.

Further, velocity of efflux is independent of the pressure gradient across BC.

Q.12 Answer:

$H/2$

Solution:

(i) If height g is h, then $\sigma_1\, g\frac{H}{3} + \sigma_2\, g\frac{H}{3} = \sigma_2 gh$

$h = \frac{1}{3}\frac{\sigma_1 + \sigma_2}{\sigma_2} H = \frac{1}{2} H$

Q.13 Answer:

$\sqrt{\frac{gH}{3}}$

Solution:

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(i) If height g is h, then $\sigma_1\ g\frac{H}{3}+\sigma_2\ g\frac{H}{3}=\sigma_2 gh$

$h=\frac{1}{3}\frac{\sigma_1+\sigma_2}{\sigma_2}H=\frac{1}{2}H$

(ii) Bernoulli's equation $P_0+\sigma_1\ g\frac{H}{3}+\sigma_2\ g\frac{2H}{3}=P_0+\sigma_2\ g\frac{2H}{3}+\frac{1}{2}\sigma_2 v^2$

$v^2=\frac{2}{\sigma_2}\cdot\sigma_1\ g\frac{H}{3}=g\frac{H}{3}=g\frac{H}{3}\quad\left.\therefore\ \frac{\sigma_1}{\sigma_2}=\frac{1}{2}\right]$

$v^2=\frac{2}{\sigma_2}\cdot\sigma_1\ g\frac{H}{3}=g\frac{H}{3}=g\frac{H}{3}$

$v=\sqrt{\frac{gH}{3}}$

(iii) $F=\sigma_2\ A_0 v^2=\sigma_2\ A_0\frac{gH}{3}$

Q.14 Answer:

$\frac{\sigma_2\ A_0 gH}{3}$

Solution:

(i) If height g is h, then $\sigma_1\ g\frac{H}{3}+\sigma_2\ g\frac{H}{3}=\sigma_2 gh$

$h=\frac{1}{3}\frac{\sigma_1+\sigma_2}{\sigma_2}H=\frac{1}{2}H$

(ii) Bernoulli's equation $P_0+\sigma_1\ g\frac{H}{3}+\sigma_2\ g\frac{2H}{3}=P_0+\sigma_2\ g\frac{2H}{3}+\frac{1}{2}\sigma_2 v^2$

$v^2=\frac{2}{\sigma_2}\cdot\sigma_1\ g\frac{H}{3}=g\frac{H}{3}=g\frac{H}{3}\quad\left.\therefore\ \frac{\sigma_1}{\sigma_2}=\frac{1}{2}\right]$

$v^2=\frac{2}{\sigma_2}\cdot\sigma_1\ g\frac{H}{3}=g\frac{H}{3}=g\frac{H}{3}$

$v=\sqrt{\frac{gH}{3}}$

(iii) $F=\sigma_2\ A_0 v^2=\sigma_2\ A_0\frac{gH}{3}$

Q.15 Answer:

$F_t=\frac{F_b}{2}$

Solution:

Principle to be used : The thrust on a plane surface whatever be its inclination, is equal to area times the height of liquid above its geometric center.

Consider element area $d\ell$ **at** ℓ **(at depth h from top)** $(h=\ell\cos\theta)$

$\therefore\quad p=h\rho g=\ell\cos\theta\rho g$

$dF=p\cdot(B\cdot d\ell)=B\rho g\cos\theta\ell d\ell$

$$F=B\rho g\cos\theta\int_0^L \ell d\ell=B\rho g\cos\theta\frac{L^2}{2}=\left(\frac{L}{2}\cos\theta\right)\rho g(LB)$$

$=$ height of column above centre $\times\ \rho g\times$ Area

$F_b=h\rho g\cdot h^2=h^3\rho g$

$$F_r=\left(\frac{h}{2}\rho g\right)\times\frac{h\cdot g}{\cos\theta}=h^3\rho g\times\frac{1}{2\cos\theta}=\frac{F_b}{2\cos\theta}$$

$$F_1=\left(\frac{h}{2}\rho g\right)(\text{ Area of trapezium })=\frac{h}{2}\rho g\left(\frac{h+\overline{h+2h\tan\theta}}{2}\right)\cdot h$$

$$=h^3\rho g\left(\frac{1+\tan\theta}{2}\right)=F_b\left(\frac{1+\tan\theta}{2}\right)$$

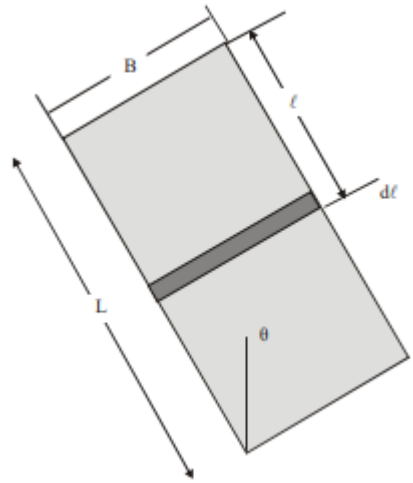
Now put various values of θ **to get required results.**

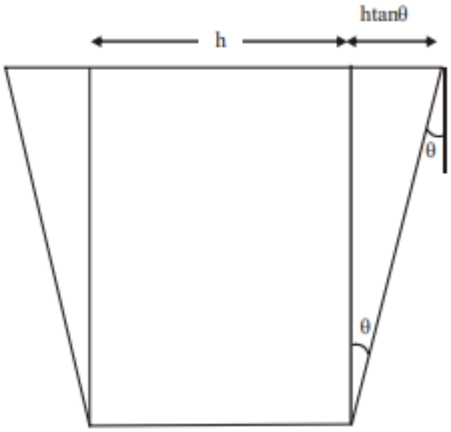
(A) $\theta=0\Rightarrow F_b, F_r=\frac{F_b}{2},\ F_t=\frac{F_b}{2}$

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(B) $\theta = 30^\circ \Rightarrow F_b, F_r = \frac{F_b}{\sqrt{3}},\ F_t = \frac{F_b}{\sqrt{3}}\frac{(\sqrt{3}+1)}{2}$

(C) $\theta = 45^\circ \Rightarrow F_b, F_r = \frac{F_b}{\sqrt{2}},\ F_t = F_b$





Q.16 Answer:

$F_r = \frac{F_b}{\sqrt{3}}$

Solution:

Principle to be used : The thrust on a plane surface whatever be its inclination, is equal to area times the height of liquid above its geometric center.

Consider element area $d\ell$ at ℓ (at depth h from top) $(h = \ell\cos\theta)$

$$\therefore \quad p = h\rho g = \ell\cos\theta\rho g$$

$$dF = p\cdot(B\cdot d\ell) = B\rho g\cos\theta\ell d\ell$$

$$F = B\rho g\cos\theta\int_0^L \ell d\ell = B\rho g\cos\theta\frac{L^2}{2} = \left(\frac{L}{2}\cos\theta\right)\rho g(LB)$$

$$= \text{ height of column above centre } \times \rho g \times \text{ Area}$$

$$F_b = h\rho g\cdot h^2 = h^3\rho g$$

$$F_r = \left(\frac{h}{2}\rho g\right)\times\frac{h\cdot g}{\cos\theta} = h^3\rho g\times\frac{1}{2\cos\theta} = \frac{F_b}{2\cos\theta}$$

$$F_1 = \left(\frac{h}{2}\rho g\right)(\text{ Area of trapezium }) = \frac{h}{2}\rho g\left(\frac{h+\overline{h+2\,h\tan\theta}}{2}\right)\cdot h$$

$$= h^3\rho g\left(\frac{1+\tan\theta}{2}\right) = F_b\left(\frac{1+\tan\theta}{2}\right)$$

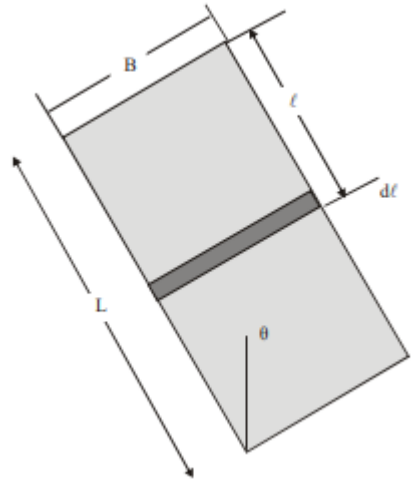
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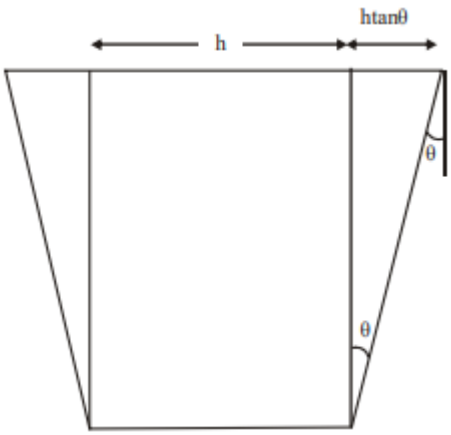
Now put various values of θ to get required results.

(A) $\theta = 0 \Rightarrow F_b, F_r = \frac{F_b}{2},\ F_t = \frac{F_b}{2}$

(B) $\theta = 30^\circ \Rightarrow F_b, F_r = \frac{F_b}{\sqrt{3}},\ F_t = \frac{F_b}{\sqrt{3}}\frac{(\sqrt{3}+1)}{2}$

(C) $\theta = 45^\circ \Rightarrow F_b, F_r = \frac{F_b}{\sqrt{2}},\ F_t = F_b$





Q.17 Answer:

$F_r = \frac{F_b}{\sqrt{2}}$

Solution:

Principle to be used : The thrust on a plane surface whatever be its inclination, is equal to area times the height of liquid above its geometric center.

Consider element area $d\ell$ at ℓ (at depth h from top) $(h = \ell\cos\theta)$

$$\therefore \quad p = h\rho g = \ell\cos\theta\rho g$$

$$dF = p\cdot(B\cdot d\ell) = B\rho g\cos\theta\ell d\ell$$

$$F = B\rho g\cos\theta\int_0^L \ell d\ell = B\rho g\cos\theta\frac{L^2}{2} = \left(\frac{L}{2}\cos\theta\right)\rho g(LB)$$

$$= \text{ height of column above centre } \times \rho g \times \text{ Area}$$

$$F_b = h\rho g\cdot h^2 = h^3\rho g$$

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$$F_r = \left(\frac{h}{2}\rho g\right) \times \frac{h \cdot g}{\cos\theta} = h^3\rho g \times \frac{1}{2\cos\theta} = \frac{F_b}{2\cos\theta}$$

$$F_1 = \left(\frac{h}{2}\rho g\right)(\text{ Area of trapezium }) = \frac{h}{2}\rho g\left(\frac{h + \overline{h + 2\,h\tan\theta}}{2}\right) \cdot h$$

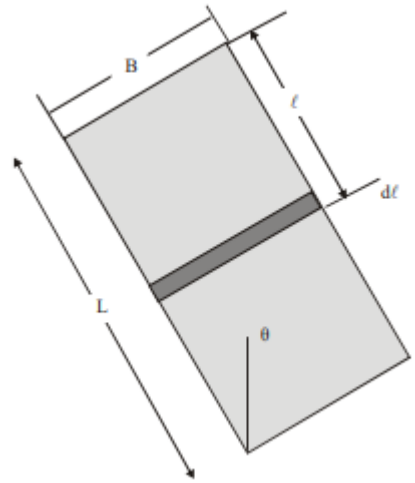
$$= h^3\rho g\left(\frac{1+\tan\theta}{2}\right) = F_b\left(\frac{1+\tan\theta}{2}\right)$$

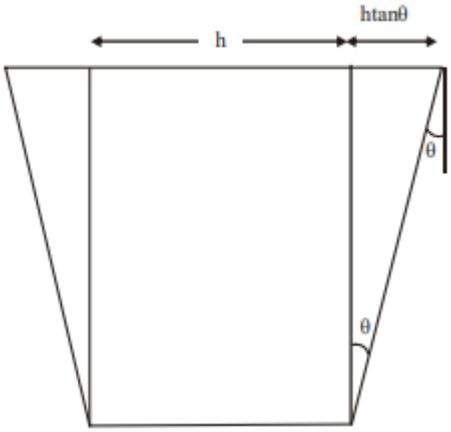
Now put various values of θ to get required results.

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(B) $\theta = 30^\circ \Rightarrow F_b, F_r = \frac{F_b}{\sqrt{3}},\ F_t = \frac{F_b}{\sqrt{3}}\frac{(\sqrt{3}+1)}{2}$

(C) $\theta = 45^\circ \Rightarrow F_b, F_r = \frac{F_b}{\sqrt{2}},\ F_t = F_b$





Q.18 Answer:

A-(qr) B-(ps) C-(qs) D-(s)

Solution:

$(A) - qr, (B) - ps, (C) - qs, (D) - s$

$h = \frac{2\,T\cos\theta}{r\rho g}$

(using $\cos 135 = -\sin 45 = -0.7$**)**

For mercury $\frac{T}{r\rho g} = 0.02$

$\therefore$ **The general formulae : For water,** $h_w = \frac{2T\cos\theta}{r\rho g} = 0.04\cos\theta$ **(metre)**

For liquid, $h_\ell = \frac{2(0.5\ T)\cos\theta}{r(0.8\rho)g} = 0.025\cos\theta$ **(metre)**

(A) $37^\circ \Rightarrow \cos\theta = 0.8 \Rightarrow h_W = 0.032\ m; h_\ell = 0.02\ m$

(B) $53^\circ \Rightarrow \cos\theta = 0.6 \Rightarrow h_W = 0.024\ m; h_\ell = 0.015\ m$

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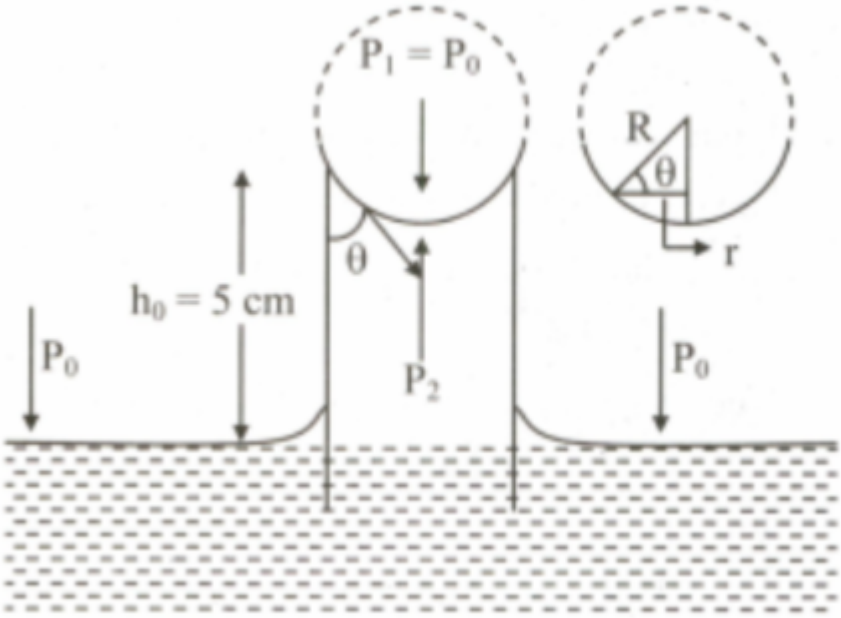
(C) $60^\circ \Rightarrow \cos\theta = 0.5 \Rightarrow h_W = 0.02\ \text{m}; h_\ell = 0.0125\ \text{m}$

(D) $90^\circ \Rightarrow$ **both zero**

Q.19 Answer:

—

Solution:



If two points are just above and below the meniscus then, $\rho g h_0 = \frac{2T}{R} \ldots\ldots (i)$

When $\theta = 0; R = r$ **and** $h_0 = h$

$\therefore \quad \rho g h = \frac{2T}{r} \ldots\ldots (ii)$

From equations (i) **and** (ii)

$\Rightarrow \frac{r}{R} = \frac{h_0}{h}$

New angle of contact,

$\theta = \cos^{-1}\left(\frac{r}{R}\right) = \cos^{-1}\left(\frac{h_0}{h}\right) = \cos^{-1}\left(\frac{5}{7.05}\right)$

$= \cos^{-1}\left(\frac{5}{5\times\sqrt{2}}\right) = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$

$\therefore \quad \theta = 45^\circ$

Q.20 Answer:

—

Solution:

Archimedes's principle applies. The buoyant force is due to the water displaced by the completely submerged life preserver and the partially submerged person.

The buoyant force is equal to the weight of the sea water displaced by the life preserver and 80% **of the person's volume.** $\rho = m/V$**. The buoyant force balances the weight of the life preserver and the person. The density of sea water is** $1030\ \text{kg/m}^3$**.**

Balancing the forces on the person-life preserver system gives

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$B = W_{\mathrm{P}} + W_{\mathrm{LP}}$. **Using** $\rho = m/V$ **gives**

$$\rho_{\mathrm{w}}\left(V_{\mathrm{LP}} + 0.800V_{\mathrm{p}}\right) = \rho_{\mathrm{p}}V_{\mathrm{p}} + \rho_{\mathrm{LP}}V_{\mathrm{LP}}.$$

For the person, we have $V_{\mathrm{p}} = m_{\mathrm{p}}/\rho_{\mathrm{p}}$ **so the last equation becomes**
$\rho_{\mathrm{w}}\left(V_{\mathrm{LP}} + 0.800m_{\mathrm{p}}/\rho_{\mathrm{p}}\right) = m_{\mathrm{p}} + \rho_{\mathrm{LP}}V_{\mathrm{LP}}$. **Solving for** ρ_{LP} **gives**
$\rho_{\mathrm{LP}} = \left[\rho_{\mathrm{w}}\left(V_{\mathrm{LP}} + 0.800m_{\mathrm{p}}/\rho_{\mathrm{p}}\right) - m_{\mathrm{p}}\right]/V_{\mathrm{LP}}$. **Putting in the numbers gives**

$$\rho_{\mathrm{LP}} = \frac{\left(1030\ \mathrm{kg/m^3}\right)\left[0.0400\ \mathrm{m^3} + \frac{(0.800)(75.0\ \mathrm{kg})}{980\ \mathrm{kg/m^3}}\right] - 75.0\ \mathrm{kg}}{0.0400\ \mathrm{m^3}} = 732\ \mathrm{kg/m^3}.$$

Evaluate: The density of the life preserver is less than that of sea water, which it must be if it is to provide buoyancy to the person in the water.

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